

SIH26103 — PRISM

Project Concept & Team Discussion Report

Purpose: This document explains only **WHAT we are building, WHY we are building it, WHO will use it, and HOW the complete monitoring workflow will behave**. Technical implementation details are intentionally kept out for now.

1. Our Core Understanding of the Problem

MoSPI has historical and ongoing information about major infrastructure projects. The team's current concern, as captured in our working notes, is that project information may reach the monitoring authority late and that reported progress, expenditure, bills, site activity and actual ground conditions may not always be easy to verify continuously. Our solution therefore combines **continuous project records, evidence-based verification, automated alerts and predictive risk intelligence** in one controlled platform.

Important: Allegations such as corruption or deliberate overbilling should not be treated as proven facts. In our project, they are treated as **risk scenarios** that the system should help detect through evidence, cross-checks and human approval.

2. What Exactly Are We Building?

PRISM — Predictive Risk & Integrated Smart Monitoring is a web-based integrated project-monitoring platform for MoSPI. It will maintain a project's journey from start to completion: project details, daily site records, photographs, geotagged inspections, materials, bills/documents, progress, delays, responsible officers and AI-generated risk signals. The platform will then use this continuously accumulated information, together with historical project data, to identify abnormal patterns, predict cost/time-overrun risk and tell the responsible MoSPI administrator where attention is required.

3. The One-Line Idea

“Create one trusted digital record of the project from Day 1 to completion, continuously verify what is being reported, and use AI to detect risk early so MoSPI can intervene before a small issue becomes a major delay or cost overrun.”

4. Complete Business / User Workflow

MoSPI Project Monitoring Administrator creates/controls a project → Project/contractor team and assigned officers submit required records → daily site evidence and documents accumulate → system checks completeness and consistency → AI identifies abnormal progress/cost/delay patterns → risk/alerts are generated → responsible officer is notified → MoSPI administrator reviews and escalates → concerned Ministry/Department/Implementing Agency takes action → action and follow-up remain recorded.

TOP-LEVEL FLOW

MoSPI → Project Monitoring Administrator → Project/Implementing Agency → Contractor → Site Team → Daily Evidence & Documents → Automated Checks + AI Risk Engine → Alerts → Administrator Review → Ministry/Department → Corrective Action → Recorded Follow-up

5. Who Controls the Platform?

MoSPI Project Monitoring Administrator is the top-level operational admin in our proposed system. The administrator controls project creation, access, monitoring visibility, alert review and escalation. Other project participants can submit information according to their role, but they do not get unrestricted control of the system.

Human-in-the-loop principle: **AI flags and recommends; the authorized human verifies and decides.**

6. Module 1 — Project Master Record

When a new project is created, the platform maintains a complete master profile instead of scattered information.

- Project name / ID and project type (road, bridge, etc.)
- Location and project scope/size
- Approved/original cost and planned duration
- Contractor / implementing agency
- Planned milestones and completion target
- Assigned officers and their responsibilities
- Other project-specific fields approved by MoSPI

Team decision: We should not hard-code every possible construction field. The project profile should have a standard core plus configurable project-specific fields.

7. Module 2 — Daily Project Record

This is the central ground-level evidence layer. Every project gets a chronological daily record so an administrator can select a date and understand what was reported that day.

- Day/date-wise work status
- Site photographs and required angles
- Physical work completed / measured progress
- Material received and/or used
- Labour/workforce information where relevant
- Machinery information where relevant
- Reason for site being inactive or work being delayed
- Officer/site visit records and remarks
- Supporting bills, receipts and documents

The key idea is **traceability**: instead of seeing only “Project = 62% complete”, the administrator can trace how the project reached 62%.

8. Photo & Evidence Workflow

The proposed system asks the uploader for context around the evidence. For example, when a first site photo is submitted, the system can ask for the date, activity, location and what happened that day. On later days it can request the information needed to maintain continuity.

Important correction: Image AI should not be described as magically knowing that a photo is “Day 1” or accurately measuring every construction quantity from any image. The safer concept is: **AI uses project**

context + submitted metadata + visual evidence to detect inconsistencies and ask targeted follow-up questions.

9. Officer Accountability & Inspection Trail

Each assigned officer gets role-specific tasks and submission requirements. The platform records whether the required inspection/response was completed, along with time, location evidence where applicable, photographs and remarks.

- Site Engineer — assigned site/progress records and visit evidence
- Sub Engineer — measurement/checking evidence when assigned
- QA/QC Engineer — quality-related inspection records
- Material Engineer — material receipt/verification records
- Billing Engineer — bills/quantity/payment-related documentation
- Planning/Project Management roles — schedule/progress review and coordination

If a required action is not completed within the configured time, the system can generate a reminder/escalation signal to the appropriate administrator.

10. Module 3 — Documentation & Verification

A separate documentation area keeps project documents, bills, receipts, approvals and inspection records in one project-linked trail. The objective is not simply storage; it is to make records cross-referenceable with the daily project record.

Example: a submitted bill can be reviewed against the corresponding work/progress record, material evidence and inspection record. **The AI may flag an inconsistency, but MoSPI-authorized personnel approve or reject it.**

11. Module 4 — Delay Intelligence

The system should capture the reason when work is delayed or a site is inactive. Instead of only showing “Delayed”, it should maintain a structured reason such as material unavailability, machinery unavailability, land/access issue, approval/clearance dependency, labour/resource issue, design/scope issue, weather or other verified category.

FLOW

Delay signal → reason captured → responsible role identified → reminder/response → administrator review → escalation/corrective action → follow-up recorded

12. Module 5 — Cost Monitoring & Overbilling Risk

The system will maintain the relationship between reported work, material records, bills and approved project information. If a bill or quantity appears inconsistent with the accumulated project record, the system raises a **verification alert** rather than automatically declaring fraud.

Example: reported physical work increases sharply, but the corresponding evidence/material/measurement records do not support the change. The platform flags the case for review. A responsible MoSPI team member then verifies it and records the decision.

13. Module 6 — Time Overrun Prediction

The platform continuously compares actual progress with the planned project trajectory. If the current trajectory indicates that the planned completion is becoming unlikely, the system generates an early warning rather than waiting until the deadline has already passed.

Example: Planned completion = March 2027. Current trajectory suggests substantial slippage. The dashboard can show a high time-overrun risk and the trend behind that risk.

14. Module 7 — Historical Data Intelligence

Historical project data becomes the learning layer. Instead of using old data only for reports, we use it to learn patterns associated with previous cost and time overruns. Current project data is then evaluated against those learned patterns.

Historical projects → learned patterns → current project data → risk probability → early warning → human review

The model should express **probability/risk**, not certainty. For example, “estimated time-overrun risk = 82%” is more appropriate than “this project will definitely be late.”

15. Module 8 — Explainable Risk

Every important risk alert should answer **WHY?** Example: risk increased because planned-vs-actual progress diverged, milestone slippage increased and expenditure progressed faster than physical work. This makes the alert actionable rather than just a red score.

16. Module 9 — Multi-Project Comparison

The administrator can compare projects by progress, planned completion, risk, cost position and other approved metrics. Because projects differ in size, sector, duration and complexity, comparisons should use comparable project groups rather than blindly ranking all projects together.

Example: “Project A is 70% complete” is less useful than “Project A is 70% complete and is behind the comparable-project trajectory.”

17. Final Integrated Workflow

PROJECT CREATION

MoSPI Admin → creates project → defines scope, cost, timeline, milestones, roles

DAILY EXECUTION

Site activity → evidence/photos → progress/material/labour/machinery details → documents/bills → officer submissions

CONTINUOUS MONITORING

Platform builds project timeline → checks missing/inconsistent records → tracks delays, visits and responsibilities

AI INTELLIGENCE

Historical patterns + current project record → cost risk + time risk + abnormality signals + explanation

ACTION

Alert → responsible officer/admin → verification → Ministry/Department escalation where needed → corrective action → follow-up recorded

18. What We Are NOT Building

To keep the team focused, the first concept should NOT become a generic project-management app, a simple CRUD dashboard, an autonomous AI decision-maker, or an automatic corruption detector. The core product is **evidence-backed monitoring + predictive risk + early warning + human verification**.

19. What We Should Improve in the Current Idea

- Use the phrase “**verification alert / anomaly alert**” instead of claiming that the AI can prove corruption.
- Keep **MoSPI administrator approval** as the final authority for sensitive verification/escalation.
- Make every important record traceable to **project** → **date** → **person/role** → **evidence** → **action**.
- Do not assume every photo can reliably provide exact quantity. Treat image analysis as supporting evidence unless validated.
- Use role-based responsibilities rather than saying “all officers can access everything.”
- Add an explicit **risk history/trend** so the administrator sees whether a project is improving or deteriorating.
- Use comparable-project benchmarking rather than simplistic all-project ranking.
- Keep the predictive module connected to the daily-record module: AI becomes stronger because the project journey is continuously recorded.

20. Jury Questions We Should Expect

Q1. Why do we need this if MoSPI already has a monitoring system?

Answer: We are proposing an intelligence and evidence layer around monitoring: a chronological daily project record, stronger traceability, cross-checking, predictive risk and early warnings. The goal is not to replace an existing government system blindly, but to strengthen decision support and monitoring.

Q2. What is the biggest problem you are solving?

Answer: The gap between reported/project-level information and timely, actionable understanding of what is actually happening on the ground—and the fact that historical data can be used more proactively to predict future risk.

Q3. Who will use the system?

Answer: MoSPI’s authorized project-monitoring administrators and relevant project/ministry/implementing-agency roles, with access controlled by responsibility.

Q4. Who has final authority?

Answer: Authorized human officers. AI generates alerts, predictions and recommendations; it does not make administrative or disciplinary decisions.

Q5. How does this reduce corruption?

Answer: We should not claim that it eliminates corruption. It increases traceability and creates cross-check opportunities between reported progress, evidence, material records, bills and inspections, making suspicious inconsistencies easier to review.

Q6. What if a contractor uploads false information?

Answer: The platform should cross-reference the submission with other evidence and records and raise inconsistencies for human verification. It should not assume that a single AI check proves truth or fraud.

Q7. What if an officer does not visit the site?

Answer: If a required visit is assigned, the system can track the expected submission and raise a reminder/escalation when evidence is missing or late. The authorized administrator decides the next action.

Q8. What is the role of historical data?

Answer: It becomes the learning layer for predictive monitoring. Historical projects teach the model patterns associated with overruns; ongoing project data is evaluated against those patterns.

Q9. What makes your idea different from a dashboard?

Answer: A dashboard mainly displays information. Our proposed platform creates a continuous evidence trail, identifies anomalies, predicts risk, explains the risk and triggers early-warning workflows.

Q10. Can AI accurately predict project delays?

Answer: It should provide an estimated probability, not certainty. Its usefulness must be evaluated on historical data with proper time-aware validation and compared with baseline methods.

Q11. Why does the system need daily records?

Answer: Because a final percentage such as 70% does not explain how the project reached that state. Daily records create a timeline that can support verification, delay analysis, accountability and prediction.

Q12. What happens after an alert?

Answer: The responsible officer/admin reviews the evidence, verifies the issue, and escalates it to the concerned authority when required. The corrective action and follow-up are recorded.

Q13. What if a project is different from historical projects?

Answer: The system should communicate uncertainty and avoid overconfident predictions. Benchmarking should use comparable projects, and human review remains essential.

Q14. Are you replacing officers with AI?

Answer: No. We are reducing repetitive monitoring and helping officers prioritize attention. Human officers remain responsible for verification and decisions.

Q15. What is your strongest innovation?

Answer: Connecting the project's continuous evidence trail with predictive risk: the same platform records what happened, detects abnormal patterns, predicts what may happen next, explains why, and routes the warning to the responsible authority.

21. Team Discussion Questions — Give Your Own Opinion Before We Freeze the Product

Your team should now debate these questions instead of immediately discussing coding:

- Which daily records are realistically possible for a site team to submit without creating excessive reporting burden?
- Which evidence should be mandatory and which should be optional?
- Which construction events should trigger an immediate alert?
- Which information can realistically be verified from photographs, and which requires a human/site measurement?
- Which roles should be allowed to submit, view, verify and approve each record?
- What should happen when the contractor and inspecting officer submit conflicting information?

- Which road/bridge-specific problems should be included as standard delay categories?
- Which project types should be benchmarked together?
- What should the administrator see first when 1,000+ projects are active?
- Which part of the idea is genuinely new compared with existing government monitoring workflows?

22. Our Current Product Definition — Freeze This for Team Discussion

PRISM is a MoSPI-controlled web platform that creates a complete, evidence-backed digital journey of infrastructure projects from Day 1 to completion, monitors daily progress and responsibilities, cross-checks project records, learns from historical projects to predict cost/time risk, explains emerging risks, and sends early warnings to authorized administrators for human verification and intervention.

Next phase: After the team agrees on this “WHAT”, we should separately design the “HOW”: exact database fields, user roles/permissions, ML targets/features, photo/evidence rules, APIs, automation triggers, UI screens and MVP scope.